# MD MAJEDUR RAHMAN

Blockchain Developer | Solidity/EVM, Solana/Rust & Web3 Backend

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Dhaka, Bangladesh



Blockchain developer with experience designing, developing, testing, and deploying **smart contracts** and full-stack blockchain systems across **Cosmos SDK/Cosmos EVM**, **Ethereum/EVM**, and **Solana**. Led most of the technical implementation across three **EtherAuthority** products: a **Solana DEX and token ecosystem** deployed to mainnet, a **React Native/Expo wallet** supporting **34 networks** and currently in pilot testing, and a **Cosmos SDK EVM-compatible blockchain** in final pre-launch testing. Built protocol, smart-contract, backend, and integration layers using **Solidity**, **Rust/Anchor**, **Go**, **TypeScript**, **Node.js**, **PostgreSQL**, **Redis**, **NATS JetStream**, and **Docker**.

## TECHNICAL SKILLS

| | |
|---|---|
| **Smart Contracts & Blockchain:** | **Solidity**, **EVM**, **Ethereum**, **Solana**, **Rust**, **Anchor**, **Cosmos SDK**, **CometBFT**, **CosmWasm**, **Hyperledger Fabric**, Go Chaincode, **IBC**, **DeFi**, ERC-20, ERC-721, ERC-1155, SPL Tokens, Token-2022, **multisig governance** |
| **Smart Contracts & Web3 Tools:** | **Hardhat**, **Foundry**, **OpenZeppelin**, **Ethers.js**, Web3.js, **Gnosis Safe**, Biconomy, Alchemy, **IPFS**, Arweave, Irys, Hyperledger Caliper |
| **Backend, APIs & Data:** | **TypeScript**, JavaScript, **Node.js**, **Go**, **Fastify**, Express.js, **REST APIs**, **gRPC**, GraphQL, Apollo Server, **WebSockets**, **PostgreSQL**, MongoDB, **Redis** (caching, pub/sub), **NATS JetStream**, LevelDB |
| **Frontend & Mobile:** | **React**, **Next.js**, **React Native**, **Redux Toolkit**, Material UI, Tailwind CSS, Vite, **Expo**, React Native Paper |
| **Infrastructure, Deploy & Security:** | **Docker**, **Kubernetes**, **Nginx**, DigitalOcean, **Grafana**, **Prometheus**, Jest, **smart contract security review**, vulnerability analysis, JWT, OAuth 2.0, biometric authentication, role-based access control |

## PROFESSIONAL EXPERIENCE

### Blockchain Developer

Dec 2025 - Jun 2026 (Full-time) | Jul 2026 - Present (Project-Based)

**EtherAuthority** | etherauthority.io — Remote

- Led most of the technical implementation across three products: a **Cosmos SDK EVM-compatible blockchain**, a **Solana DEX and token ecosystem**, and a **multi-chain crypto wallet mobile application**.
- Developed and deployed smart contracts and blockchain programs across **Cosmos SDK/Cosmos EVM**, **Solidity/EVM**, and **Solana Anchor/Rust**, with deployments to devnet, testnet, and mainnet.
- Designed and implemented **Go**, **TypeScript**, **Node.js**, and **Fastify** backend services and REST APIs backed by **PostgreSQL**, **Redis**, **NATS JetStream**, **Nginx**, and **Docker**.
- Built Solana DEX and token-launchpad infrastructure with a **TypeScript SDK**, **Anchor programs**, **Orca Whirlpools** and **Raydium** integrations, **Squads v4 multisig governance**, and automated liquidity workers.
- As a project-based engagement, independently delivered the complete crypto wallet project across **Android**, **iOS**, and a **Chrome extension**, supporting **34 networks**, user-configurable network addition, **EVM and Solana assets**, ERC-20 and SPL token transfers, NFTs, **biometric unlock**, auto-lock protection, and multi-account management; the wallet is currently in pilot testing.
- Contributed to **React** and **React Native** UI development and integration for blockchain-based applications.
- Performed **smart contract security reviews** and **vulnerability analysis** across blockchain applications.

### Co-Founder & CTO

Feb 2024 - Present | Pre-launch development

**Patient Bio (Early-Stage Startup)** | patientbio.app — Full-time

- Lead product architecture and development for an early-stage **decentralized healthcare platform** currently in pre-launch development.
- Translate product requirements into **technical architecture**, **engineering practices**, and an **execution roadmap**.
- Lead development from **research and prototyping** through **product implementation** and **launch readiness**.
- Manage a distributed team and coordinate engineering execution **across product development initiatives**.

## SELECTED BLOCKCHAIN PROJECTS

### Sovereign — Cosmos SDK EVM-Compatible Blockchain | LINK

Independent Lead Developer | Final testing; preparing for launch

- Owned the end-to-end design and implementation of the blockchain project, covering **protocol architecture**, backend services, infrastructure, testing, and application integration.
- Built a **Cosmos SDK blockchain** combining **CometBFT consensus**, **Cosmos EVM compatibility**, **CosmWasm smart contracts**, **IBC interoperability**, and **dual Solidity/Rust execution**.
- Designed and implemented custom modules for **validator certification**, **validator power equalization**, **price oracles**, **milestones**, **settlement**, **bridging**, **governance extensions**, **ERC-20 bridging**, and **EIP-1559 fees**.
- Built a **BSC bridge** with **Solidity LockBox escrow**, **Gnosis Safe governance**, a **Go relayer**, **threshold signing**, and **circuit-breaker controls**.
- Developed **CQRS backend services** and **REST/gRPC APIs**, with **NATS JetStream messaging** and **PostgreSQL/Redis data pipelines**.
- Deployed chain nodes and supporting services with **Kubernetes Helm charts** and **raw Kubernetes manifests**; configured **Prometheus metrics**, **Grafana dashboards**, alert rules, and **HashiCorp Vault secret management** for chain health and bridge operations.
- Measured an average **block production time of approximately 1 second** and **transaction confirmation time of approximately 1.3 seconds** in benchmark testing.
- Built a custom **45+ page blockchain explorer** with a **Go real-time indexer**, REST/gRPC APIs, Redis caching, **PgBouncer connection pooling**, and multi-wallet integrations including **Keplr, MetaMask, Cosmostation, and OKX Wallet**.
- Implemented a **devnet token faucet service** and created **34 end-to-end test files** covering core blockchain and application workflows.

**Tech:** Go, Cosmos SDK, CometBFT, Cosmos EVM, Rust, CosmWasm, Solidity, Foundry, IBC, TypeScript, NATS JetStream, PostgreSQL, TimescaleDB, Redis, PgBouncer, Docker, Kubernetes, Prometheus, Grafana, HashiCorp Vault

### DeFiApps — Multi-Module DeFi Ecosystem | LINK

Sepolia Testnet

- Built a full-stack DeFi platform featuring an **AMM for token swaps with low slippage**, **liquidity provision** (earning swap fees), **dynamic yield farming**, **staking**, and **LP reward harvesting**.
- Implemented **lending and borrowing** for over-collateralized loans with **price-oracle valuation**, **dynamic health-factor monitoring**, and **liquidation-risk signals**.
- Deployed four Solidity smart contracts to Ethereum Sepolia and integrated a **2-of-3 Gnosis Safe multisig vault** for **protocol treasury security**, on-chain proposal approvals, and multi-party signing.
- Developed high-performance **Fastify APIs with JSON-RPC optimization**, **Redis caching**, **real-time on-chain event listeners**, and **React 18 dashboards** for unified position and treasury analytics.

**Tech:** Solidity 0.8.24, Hardhat, OpenZeppelin, Ethers.js v6, Gnosis Safe, Fastify, Redis, React 18, Material UI

### Solana DEX, Token Launchpad & Liquidity Ecosystem | LINK

Deployed to Solana mainnet

- Built an end-to-end **Solana DEX**, token launchpad, **Squads v4 governance**, and automated liquidity ecosystem using **Anchor/Rust smart contracts** and a **TypeScript SDK**.
- Implemented **versioned transactions** with **Address Lookup Tables**, configurable priority fees, **WebSocket signature confirmation**, and **Jito bundle submission**.
- Built a **Fastify REST API** with **PostgreSQL**, **Redis caching**, and background workers for swaps, token creation, liquidity, price feeds, metrics, governance, referrals, and automated fee collection.
- Integrated **Orca Whirlpools**, **Raydium pools**, **Irys and Arweave metadata storage**, and **Solana Token-2022 functionality**; deployed the system to **Solana mainnet**.

**Tech:** Solana, Anchor, Rust, TypeScript, Node.js, SPL Tokens, Token-2022, Orca Whirlpools, Program Derived Addresses (PDAs), Fastify, PostgreSQL, Redis, Docker

### NFT Nexus — Multi-Chain Real-Time NFT Marketplace | LINK

Solana Devnet & Sepolia

- Built a **multi-chain NFT marketplace** supporting discovery, portfolio management, analytics, trading, resales, and administration workflows.
- Integrated **Solana Devnet** using **Anchor/Rust** and **EVM Sepolia** with **creator royalties** and **marketplace owner fees**.
- Implemented real-time updates using **NATS JetStream**, **Fastify WebSockets**, **indexer backfill and lookback processing**, signature idempotency, **IPFS/Pinata**, and **Arweave/Irys media storage**.

**Tech:** Solidity, Rust, Anchor, Foundry, React 18, Vite, Fastify, NATS JetStream, PostgreSQL, Redis, WebSockets, IPFS, Arweave

### Custom Multi-Platform Crypto Wallet | LINK

Pilot testing

- Independently delivered the complete crypto wallet project for **Android, iOS, and a Chrome extension** using **React Native and Expo**, supporting **34 networks**, EVM and Solana assets, user-configurable network addition through Settings, wallet creation and import, multi-account management, native asset transfers, **ERC-20 and SPL token transfers**, and **NFT support**.
- Implemented **biometric unlock**, password fallback, configurable auto-lock protection, and persistent local state using **Redux Persist** and **AsyncStorage**.

**Tech:** React Native, Expo, TypeScript, Redux Toolkit, Redux Persist, AsyncStorage, Expo Router, Expo Local Authentication, Styled Components, Sentry, EVM, Ethereum, Solana

## AWARDS & RECOGNITION

- **Top 15** — Startup World Cup, Las Vegas, USA | 2026
- **Award of Merit** — International Blockchain Olympiad (IBCOL) | 2024
- **Finalist and Honorable Mention** — Bangladesh Blockchain Olympiad | 2023, 2024
- **1st Runner-Up** — Spark Business Competition | 2024
- **2nd Runner-Up** — Business Idea Competition | 2023

## EDUCATION

**Bachelor of Science in Computer Science and Engineering** — Daffodil International University | April 2021 - May 2025 | **CGPA: 3.33/4.00**